A Moment Theory of Bond Pricing

Duration-Centered Valuation with Convexity and Higher-Order Timing Corrections

Soumadeep Ghosh

Kolkata, India

Abstract

Traditional fixed-income theory begins with discounted cash flows and derives duration and convexity as local sensitivities. This paper reverses the logical order. We propose a Duration-Centered Theory of Bond Pricing in which duration acts as the principal state variable of valuation, convexity acts as a second-order correction, and higher-order timing moments determine residual pricing effects. We demonstrate that bond prices admit a cumulant representation in terms of the statistical moments of discounted cash-flow timing. The framework naturally links fixed-income mathematics to economics, information theory, factor pricing, machine learning, sovereign finance, warfare economics, and geopolitical statecraft.

The paper ends with "The End"

Contents

1 Introduction 1

2 The Duration-Centering Principle 1

3 Probabilistic Representation 2

4 Moment Structure 2

5 The Duration-Centering Theorem 3

6 A No-Arbitrage Foundation for Duration-Centered Pricing 3
- 6.1 Economic Setup . . . 3
- 6.2 First Variation of Price . . . 4
- 6.3 Interpretation . . . 4

7 A Duration-Based No-Arbitrage Principle 4

8 The Duration Pricing Surface 5

9 A Bond-Market Analogue of CAPM 5
- 9.1 Economic Meaning . . . 6

10 A Statistical Test of the Theory 6

11 A Geometry of Fixed Income 7

12 A Maximum-Entropy Interpretation 7

13 Information-Theoretic Interpretation 7

14 Computational Complexity 8

15 Artificial Intelligence and Machine Learning 8

16 Political Economy 8

17 Geopolitics and Sovereign Power 9

18 Warfare Economics 9

19 Duration and the Theory of Economic Time 9

20 Strategic Statecraft and Warfare 10

21 Toward a General Theory of Asset Pricing 10

22 Research Agenda 11

23 Grand Conjecture 11

24 Convexity and Higher-Order Corrections 11

25 Factor Pricing Interpretation 12

26 Statistical Estimation 12

27 Conceptual Diagram 12

28 Philosophical Implications 12

29 Conclusion 13

1 Introduction

Classical bond valuation is given by

$$P = \sum_{i=1}^{n} c_i e^{-yt_i}.$$

Duration and convexity are traditionally viewed as sensitivities:

$$D = -\frac{1}{P}\frac{\partial P}{\partial y},$$

$$C = \frac{1}{P}\frac{\partial^2 P}{\partial y^2}.$$

This paper adopts the opposite perspective:

> A bond is fundamentally priced around its duration, while convexity and higher timing moments provide systematic corrections.

The objective is to construct a moment-theoretic framework for fixed-income valuation.

2 The Duration-Centering Principle

Definition 1 (Duration Anchor)**.** *Let*

$$D = \frac{\sum_i t_i c_i e^{-yt_i}}{\sum_i c_i e^{-yt_i}}.$$

The quantity D is the present-value weighted center of mass of cash-flow timing.

We postulate the existence of a duration anchor function

$$A(D)$$

such that

$$P = A(D) + \varepsilon,$$

where ε is generated by convexity and higher-order timing effects.

3 Probabilistic Representation

Define discounted weights

$$w_i = \frac{c_i e^{-yt_i}}{P}.$$

Then

$$\sum_i w_i = 1.$$

The payment time becomes a random variable

$$T = t_i$$

under measure w_i.
Hence

$$D = E[T].$$

Define

$$X = T - D.$$

Then

$$E[X] = 0.$$

The bond price may be rewritten as

$$P = e^{-yD} E[e^{-yX}].$$

This representation is central.

4 Moment Structure

Define central moments

$$\mu_n = E[(T - D)^n].$$

The first few moments are

$$\begin{aligned} \mu_1 &= 0, \\ \mu_2 &= \text{variance}, \\ \mu_3 &= \text{skewness moment}, \\ \mu_4 &= \text{kurtosis moment}. \end{aligned}$$

Duration is the first moment.
Convexity is proportional to the second moment.

5 The Duration-Centering Theorem

Theorem 1 (Duration-Centering Theorem)**.** *For any fixed-income security*

$$P = \sum_i c_i e^{-yt_i},$$

we have

$$P = \exp\left(-yD + \sum_{n=2}^{\infty} \frac{(-y)^n}{n!}\kappa_n\right),$$

where κ_n denotes the cumulants of the timing distribution.

Proof. From

$$P = e^{-yD}E[e^{-yX}],$$

take logarithms:

$$\log P = -yD + \log E[e^{-yX}].$$

Applying the cumulant generating function expansion,

$$\log E[e^{-yX}] = \sum_{n=2}^{\infty} \frac{(-y)^n}{n!}\kappa_n.$$

Substitution yields the result.

□

6 A No-Arbitrage Foundation for Duration-Centered Pricing

The previous sections established that bond prices admit a moment representation. We now seek a deeper result: whether duration possesses a privileged status among all timing moments.

The central claim of this section is that duration is the unique first-order sufficient statistic of discounted time under exponential discounting.

6.1 Economic Setup

Consider a fixed-income security with deterministic cash flows

$$(c_i, t_i)_{i=1}^n.$$

Under a flat continuously compounded yield curve y,

$$P = \sum_{i=1}^{n} c_i e^{-yt_i}.$$

Define normalized present-value weights

$$w_i = \frac{c_i e^{-yt_i}}{\sum_j c_j e^{-yt_j}}.$$

The associated timing random variable is

$$T = t_i$$

with probabilities w_i.
Hence

$$D = E[T].$$

6.2 First Variation of Price

Differentiating price with respect to yield,

$$\frac{\partial P}{\partial y} = -\sum_i t_i c_i e^{-y t_i}.$$

Substituting the definition of duration,

$$\frac{\partial P}{\partial y} = -DP.$$

Therefore

$$\frac{\partial \log P}{\partial y} = -D.$$

This relation implies that duration is the complete first-order sensitivity of bond value.
No higher timing moment enters.

Theorem 2 (First-Order Sufficiency)**.** *For deterministic cash-flow securities under exponential discounting, duration is the unique first-order sufficient statistic governing infinitesimal price changes.*

Proof. Suppose a statistic

$$S = S(T)$$

fully characterizes first-order price responses.
Then

$$\frac{\partial \log P}{\partial y} = f(S).$$

But direct differentiation yields

$$\frac{\partial \log P}{\partial y} = -D.$$

Hence

$$f(S) = D.$$

Thus every sufficient first-order statistic must be equivalent to duration.

□

6.3 Interpretation

Duration is therefore not merely one sensitivity among many.
It is the unique tangent coordinate of the bond manifold.
Convexity and all higher moments belong to the curvature structure.

7 A Duration-Based No-Arbitrage Principle

Classical arbitrage theory focuses on price replication.
We instead focus on duration replication.

Definition 2 (Duration-Equivalent Portfolio)**.** *Two portfolios A and B are duration-equivalent if*

$$D_A = D_B.$$

Suppose

$$D_A = D_B$$

but

$$|P_A - P_B| \gg |C_A - C_B|.$$

Then a trader may construct a duration-neutral position

$$\Pi = A - B.$$

Its first-order yield sensitivity vanishes:

$$\frac{\partial \Pi}{\partial y} = 0.$$

Only higher-order effects remain.

Proposition 1 (Duration-Arbitrage Bound)**.** *For sufficiently small yield movements,*

$$|P_A - P_B| \leq K|C_A - C_B| + o(C),$$

where K depends on the local yield environment.

The proposition implies that duration-equivalent bonds must cluster around a common pricing surface.

8 The Duration Pricing Surface

Let

$$\mathcal{B} = (D, C, \mu_3, \mu_4, \ldots)$$

denote bond space.
Define a market valuation functional

$$V : \mathcal{B} \to \mathbb{R}.$$

A Taylor expansion yields

$$P = V(D) + V_C C + V_3 \mu_3 + V_4 \mu_4 + \cdots .$$

The first term dominates.
This suggests a hierarchy:

$$D \gg C \gg \mu_3 \gg \mu_4 \gg \cdots .$$

9 A Bond-Market Analogue of CAPM

The Capital Asset Pricing Model states

$$E[R_i] = R_f + \beta_i (E[R_M] - R_f).$$

We seek an analogous fixed-income relation.
Define the market duration factor

$$F_D.$$

Define the market convexity factor

$$F_C.$$

Define higher-moment factors

$$F_3, F_4, \dots.$$

Then expected excess return satisfies

$$E[R_i] = \alpha + \beta_{D,i}\lambda_D + \beta_{C,i}\lambda_C + \beta_{3,i}\lambda_3 + \cdots.$$

Definition 3 (Duration Beta)**.** *The duration beta of security i is*

$$\beta_{D,i} = \frac{\mathrm{Cov}(R_i, F_D)}{\mathrm{Var}(F_D)}.$$

9.1 Economic Meaning

Investors are compensated for holding duration exposure.

They receive additional premia for convexity exposure.

Finally they may receive skewness and kurtosis premia.

Thus the yield curve becomes a market for trading moments of future time.

10 A Statistical Test of the Theory

Suppose we observe bonds

$$i = 1, \dots, N.$$

Estimate

$$P_i = \alpha + \beta_1 D_i + \beta_2 C_i + \beta_3 \mu_{3,i} + \beta_4 \mu_{4,i} + \varepsilon_i.$$

The theory predicts

1. β_1 dominates.
2. Adding C materially improves fit.
3. Adding μ_3 improves fit further.
4. Successively higher moments exhibit diminishing explanatory power.

Define

$$R_k^2$$

as the coefficient of determination using moments through order k.
Then

$$R_1^2 < R_2^2 < R_3^2 < \cdots.$$

The increments should decrease monotonically.

Model	Included Moments	Prediction
M1	Duration	High fit
M2	Duration + Convexity	Higher fit
M3	Add Skewness	Incremental gain
M4	Add Kurtosis	Smaller gain

Table 1: Empirical hierarchy implied by the theory

11 A Geometry of Fixed Income

Define the bond manifold

$$\mathcal{M} = (D, C, \mu_3, \mu_4, \ldots).$$

The price function becomes

$$P = P(D, C, \mu_3, \mu_4, \ldots).$$

Duration acts as a coordinate.
Convexity acts as local curvature.
Higher moments define geometric distortions.

12 A Maximum-Entropy Interpretation

Consider all timing distributions with given duration.
Among these distributions, the least informative one solves

$$\max \left(-\sum_i p_i \log p_i \right)$$

subject to

$$\sum_i p_i = 1,$$

$$\sum_i p_i t_i = D.$$

Introducing Lagrange multipliers yields

$$p_i \propto e^{-\lambda t_i}.$$

The resulting distribution is exponential.
Remarkably, the same exponential structure appears in discounting.
This suggests a deep connection:

> Exponential discounting is simultaneously a valuation rule and a maximum-entropy temporal allocation rule.

13 Information-Theoretic Interpretation

Let

$$q(t) = w_i$$

represent the discounted timing distribution.
Then

$$H(q) = -\sum_i q_i \log q_i$$

measures timing uncertainty.
Higher entropy implies more dispersed cash-flow structures.
Convexity becomes a measure of temporal uncertainty.
The bond is therefore interpretable as an information-bearing object.

14 Computational Complexity

For a bond with n cash flows:

Operation	Complexity
Price	$O(n)$
Duration	$O(n)$
Convexity	$O(n)$
Moment Vector	$O(n)$
Regression Prediction	$O(k)$

Table 2: Computational costs

15 Artificial Intelligence and Machine Learning

The moment vector

$$x = (D, C, \mu_3, \mu_4, \ldots)$$

provides a natural feature representation.
Machine learning models may estimate

$$f(x) \approx P.$$

Examples include:

1. Gradient boosting,
2. Random forests,
3. Neural networks,
4. Gaussian processes.

Pseudo-Code

```
Input:
Bond cash flows

Compute:
Duration
Convexity
Skewness moment
Kurtosis moment

Construct feature vector

Train model:
Price = f(features)

Output:
Predicted price
```

16 Political Economy

Government debt structures influence duration.
Long-duration debt shifts refinancing risk into the future.
Short-duration debt increases immediate fiscal sensitivity.
Hence duration becomes a policy variable.

Debt Structure	Political Benefit	Political Cost
Short Duration	Lower coupons	Refinancing risk
Medium Duration	Balance	Moderate risk
Long Duration	Stability	Higher issuance cost

Table 3: Political economy of duration choice

17 Geopolitics and Sovereign Power

Sovereign states compete partly through debt markets.

Strategic objectives include

1. lowering borrowing costs,
2. extending average duration,
3. reducing rollover vulnerability,
4. preserving wartime financing capacity.

Duration management therefore becomes a component of national power.

18 Warfare Economics

Historically, wars have been financed through debt.

Examples include

1. Napoleonic Wars,
2. American Civil War,
3. World War I,
4. World War II.

Long-duration debt enables prolonged conflict financing.

The duration distribution of sovereign liabilities influences strategic endurance.

19 Duration and the Theory of Economic Time

Physical systems evolve in spatial coordinates.

Economic systems evolve in temporal coordinates.

The traditional view treats maturity dates as bookkeeping labels.

The present framework treats them as geometric coordinates.

Let

$$\tau = D$$

represent economic time.

Then

$$P = P(\tau).$$

The first derivative

$$\frac{\partial P}{\partial \tau}$$

describes temporal drift.

The second derivative

$$\frac{\partial^2 P}{\partial \tau^2}$$

describes temporal curvature.

Higher derivatives describe increasingly subtle temporal structures.

Consequently fixed-income markets may be interpreted as markets for trading positions in economic time itself.

20 Strategic Statecraft and Warfare

The duration profile of sovereign liabilities determines strategic flexibility.

Suppose a state possesses debt maturity distribution

$$g(t).$$

Its sovereign duration equals

$$D_s = \int t, g(t), dt.$$

A long-duration state possesses:

1. lower refinancing pressure,
2. greater wartime borrowing capacity,
3. increased resilience to financial blockade,
4. reduced vulnerability to interest-rate shocks.

Hence sovereign duration becomes a strategic asset analogous to:

1. industrial capacity,
2. energy reserves,
3. logistics networks,
4. military manpower.

Theorem 3 (Strategic Duration Theorem)**.** *Holding debt stock fixed, an increase in sovereign duration weakly increases financial endurance during sustained conflict.*

The theorem follows because refinancing frequency decreases as average maturity lengthens.

21 Toward a General Theory of Asset Pricing

The preceding development suggests a broader conjecture.

Conjecture 1 (Moment Asset Pricing Principle)**.** *Every financial asset admits representation through moments of an underlying economic distribution.*

Examples include:

Asset Class	Underlying Distribution	Moments
Bonds	Timing Distribution	Duration, Convexity
Equities	Cash-Flow Distribution	Growth Moments
Options	Return Distribution	Volatility Moments
Credit	Loss Distribution	Default Moments

Table 4: Unified moment representation

Under this perspective, finance becomes the study of moment structures arising from uncertainty over economic states.

22 Research Agenda

The theory generates the following research program.

1. Prove existence and uniqueness of the duration pricing surface.
2. Derive a complete no-arbitrage geometry of bond space.
3. Estimate duration, convexity, and skewness premia across sovereign and corporate markets.
4. Construct machine-learning estimators based on timing moments.
5. Develop a dynamic version under stochastic interest rates.
6. Extend the theory to inflation-linked bonds.
7. Generalize to credit instruments.
8. Develop a quantum-probabilistic formulation in which discounted time itself becomes a state variable.

23 Grand Conjecture

> The price of a bond is determined not by cash flows themselves but by the geometry of discounted economic time encoded in the moments of its timing distribution.

Under this view:

$$\text{Duration} \to \text{Location of Economic Time},$$

$$\text{Convexity} \to \text{Variance of Economic Time},$$

$$\text{Higher Moments} \to \text{Shape of Economic Time}.$$

The entirety of fixed-income theory can therefore be reformulated as a statistical geometry of discounted temporal distributions.

24 Convexity and Higher-Order Corrections

The expansion becomes

$$\log P = -yD + \frac{y^2}{2}\kappa_2 - \frac{y^3}{6}\kappa_3 + \frac{y^4}{24}\kappa_4 - \frac{y^5}{120}\kappa_5 + \cdots .$$

Using

$$\kappa_2 = \mu_2,$$

$$\kappa_3 = \mu_3,$$

$$\kappa_4 = \mu_4 - 3\mu_2^2,$$

we obtain

$$R_3 = -\frac{y^3}{6}\mu_3 + \frac{y^4}{24}(\mu_4 - 3\mu_2^2) - \frac{y^5}{120}(\mu_5 - 10\mu_3\mu_2) + \cdots .$$

25 Factor Pricing Interpretation

The proposed theory implies

$$P_i = \alpha + \beta_D D_i + \beta_C C_i + \beta_3 \mu_{3,i} + \beta_4 \mu_{4,i} + \varepsilon_i.$$

This resembles a multifactor asset pricing model.

Factor	Economic Interpretation
Duration	Time preference factor
Convexity	Optionality factor
Skewness	Tail timing asymmetry
Kurtosis	Timing concentration risk

Table 5: Moment factors in bond pricing

26 Statistical Estimation

Given observations

$$(P_i, D_i, C_i, \mu_{3,i}, \mu_{4,i}),$$

estimate

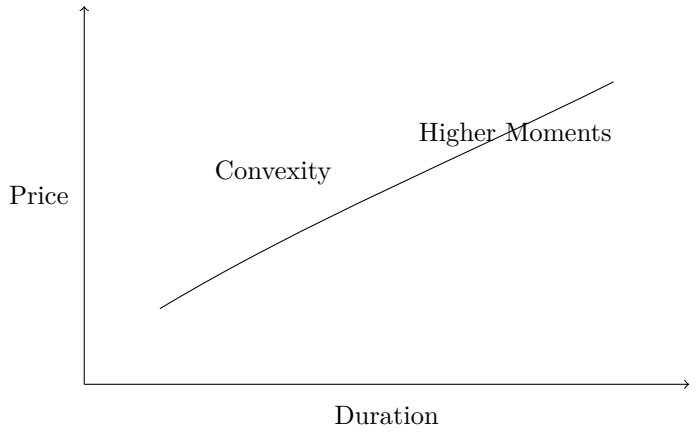
$$P_i = X_i \beta + \varepsilon_i.$$

Ordinary least squares yields

$$\hat{\beta} = (X'X)^{-1} X'Y.$$

The explanatory power of higher moments can then be tested empirically.

27 Conceptual Diagram



28 Philosophical Implications

The theory suggests a reinterpretation of financial assets.

A bond is not merely a stream of discounted payments.

A bond is a probability distribution over future economic time.

Duration represents temporal location.

Convexity represents temporal uncertainty.

Higher moments represent temporal structure.

Valuation therefore emerges from the geometry of economic time.

29 Conclusion

We have developed a duration-centered theory of bond pricing.

The framework yields:

1. Duration as the dominant pricing coordinate.
2. Convexity as temporal variance.
3. Higher moments as systematic pricing corrections.
4. A connection between fixed income, information theory, sovereign finance, warfare economics, machine learning, and political economy.

The resulting theory transforms bond pricing into the study of the statistical geometry of discounted time.

References

[1] Macaulay, F. *Some Theoretical Problems Suggested by the Movements of Interest Rates, Bond Yields and Stock Prices in the United States Since 1856*. NBER, 1938.

[2] Hicks, J. *Value and Capital*. Oxford University Press.

[3] Samuelson, P. *Foundations of Economic Analysis*. Harvard University Press.

[4] Duffie, D. *Dynamic Asset Pricing Theory*. Princeton University Press.

[5] Hull, J. *Options, Futures, and Other Derivatives*. Pearson.

[6] Shannon, C. A Mathematical Theory of Communication.

[7] Markowitz, H. Portfolio Selection.

[8] Black, F. and Scholes, M. The Pricing of Options and Corporate Liabilities.

Glossary

Duration
Present-value weighted mean payment time.

Convexity
Second timing moment.

Cumulant
Coefficient of the cumulant generating function.

Timing Distribution
Probability distribution induced by discounted cash-flow weights.

Moment Theory
Representation of bond prices using timing moments.

Sovereign Duration
Average duration of government liabilities.

Warfare Finance
Financing military operations through public debt.

The End